

SEFire Lab 1 Outline

Team Sapphire

CS 410

21 Sep 2026

Participants: Justin Jude Javier, Jared Chavira, Alexis Adams, Luke Gibson, Keith Hamlin,

Laurie Kreger, Sully Mjed, Kevin Neri

TABLE OF CONTENTS

1. Introduction.....	3
1.1 The Societal Problem	3
2. SEFire Product Description.....	4
2.1 Key Product Features and Capabilities.....	4
2.2 Major Components (Hardware/Software).....	5
3. Identification of Case Study.....	6
4. Glossary.....	7
5. References.....	8

List of Figures

Figure 1: The difficulty in covering expenses and paying bills.....	3
---	---

1. Introduction

1.1 The Societal Problem

Managing financial responsibilities in shared households can be difficult.

Tracking who is responsible for rent, utilities, groceries, and other shared expenses can become confusing and may lead to missed payments or disagreements. In a 2024 Survey conducted by the Financial Industry Regulatory Authority (FINRA), 44% of the respondents reported that it was not at all difficult to cover expenses or pay bills (Lin et al., 2025). In contrast, this implies that 56% of respondents found it difficult to pay their bills. Money can also be a source of conflict between roommates. Of 1,257 U.S. roommates surveyed by SpareRoom, 26% of disputes were related to money or utility bills (SpareRoom, 2025).

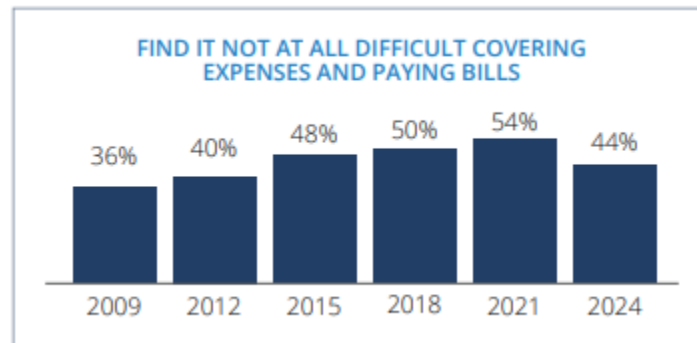


Figure 1: The difficulty in covering expenses and paying bills

The solution is an application designed to assist with tracking expenses, as well as dividing them as necessary among roommates or household members. SEFire is a new application that keeps all shared expenses in one place. Users can then choose who is supposed to pay for what. They can always look at the updated balances and also note when payment has been made.

2. SEFire Product Description

SEFire is a web application that allows roommates, couples, and families an easier way to manage the money that they spend together as a household. Instead of tracking expenses through memory, text messages, or verbal agreements, users can track and log expenses, choose how to split payments between different household members, and see who still has payments due.

Receipts can be uploaded and payments marked as fulfilled, so that managing expenses can be less manual and mentally taxing. SEFire is more than just an expense management application; it also provides financial and budgeting vocabulary definitions to better users' understanding of financial literacy. SEFire aims to reduce the confusion, stress, and manual overhead that shared expenses often create and give users tools to better understand and manage their finances.

2.1. Key Product Features and Capabilities

SEFire allows users to add expenses, choose how they are divided among household members, including dividing them into even or custom amounts, view the owed balances, and mark payments as complete. Users can also upload photos of receipts and extract expense details with optical character recognition (OCR) processing. Users can also view and correct the extracted information before it is saved. To set SEFire apart from other expense management tools, the application provides explanations and definitions of financial vocabulary and tools for setting savings goals, such as saving for a shared trip. By combining expense tools with financial literacy education, SEFire addresses both causes of the societal problem of the practical difficulty of tracking expenses and the lack of financial literacy that makes budgeting for expenses difficult.

2.2. Major Components (Hardware/Software)

2.2.1 *Hardware:*

2.2.1.1 User Devices: The application will primarily use existing consumer hardware, i.e a smartphone, mobile device, or tablet.

2.2.1.2 Hosting Infrastructure: Cloud servers and storage supporting Vercel and Supabase will host the web application, backend services, database, and receipt files.

2.2.2 *Software:*

2.2.2.1 Web/App Interface: React Native will be used to develop the application's interface, display it in web browsers across different devices, and allow users to interact with it.

2.2.2.2 Backend/Application Server: Processes requests, performs calculations, and communicates with Supabase and the PostgreSQL database.

2.2.2.3 Database and Storage: PostgreSQL will store household membership, expenses, splits, payment records, and savings goals. Supabase Storage will store receipt images linked to expense records.

2.2.2.4 OCR Processing: Tesseract.js. Is proposed to run Tesseract OCR in the web browser. It will extract text into suggested expense fields for users to review and correct before saving.

2.2.2.5 Authentication and Security: Protects user accounts from unauthorized access and ensures that only users can access their own financial information.

2.2.2.6 Financial Management Features: Provide transaction tracking, budgeting, spending analysis, and financial summaries.

Together, these components form the overall architecture of the finance application SEFire.

3. Identification of Case Study

3.1 WHO: Robert, a 21-year-old who is rooming with two friends, Jerry, a 22-year-old, and Bob, a 23-year-old, in an apartment. They split expenses such as rent, utilities, subscriptions, and groceries.

3.2 PROBLEM: Robert and his roommates need a clear way to track who paid each bill, how much each person owes, and which payments have been completed.

3.3 FEATURES: Entering expenses, custom splitting, and marking completed payments.

3.4 SCENARIO: Robert pays a \$100 electricity bill for the apartment and enters the expense into SEFire, which is then divided into percentages based on custom splitting. They agreed on a 44/34/22 split. SEFire assigns \$44 to Robert, who made the initial purchase, is marked paid. SEFire shows Jerry owes \$34 and Bob \$22. Upon payment, SEFire marks them as paid, and the request is archived.

3.5 FUTURE: SEFire would be spread by word of mouth to friends and neighbors who regularly divide expenses.

4. Glossary

API — Application Programming Interface

Git — Distributed version control system

GitHub — Software development and repository-hosting platform

MFCD — Major Functional Component Diagram

OCR — Optical Character Recognition

PostgreSQL — Relational database management system

React Native — Framework for developing application interfaces

React Native for Web — Technology for using React Native components in web browsers

SEFire — Shared household Expense management application

Supabase — Backend development platform

Tesseract OCR — Optical character recognition engine

UI — User interface, which is the point of contact and interaction between a human and a device, program, or website.

Vercel — Web application hosting and deployment platform

VS Code — Visual Studio Code

5. References

SpareRoom. (2025). *How to avoid roommate wars*. Spareroom.Com.

<https://www.spareroom.com/content/info-statistics/avoid-roommate-disputes/>

Lin, J. T., Bumcrot, C., Lusardi, A., Valdes, O., Mottola, G., Ganem, R., Sarver, S., Kieffer, C., McLaughlin, R., & Walsh, G. (2025). *Financial Capability in the United States: Results from the FINRA Foundation's National Financial Capability Study (6th Edition)*. FINRA Investor Education Foundation.

<https://finrafoundation.org/sites/finrafoundation/files/2025-07/NFCS-Report-Sixth-Edition-July-2025.pdf>